Data

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x	y	x^2	y^2	xy
19	29	361	841	551
25	23	625	529	575
23	26	529	676	598
22	27	484	729	594
89	105	1999	2775	2318

$$r = \frac{4\times 2318 - 89\times 105}{\sqrt{(4\times 1999 - 89^2)(4\times 2775 - 105^2)}}$$

$$= -0.973$$

$$b_{x/y} = \frac{4\times 2318 - 89\times 105}{4\times 2775 - 105^2} = -0.973$$

$$b_{y/x} = \frac{4\times 2318 - 89\times 105}{4\times 1999 - 89^2} = -0.973$$

Section-E

1.

$r = -0.973$

$b_{x/y} = -0.973$

RL, $\hat{x} = -0.973\left(y - \frac{105}{4}\right) + \frac{89}{4}$

$\Rightarrow \hat{x} = -0.973y + 47.791$

$\hat{x}(25) = 23.47$

2. $r = -0.8$, the variables are change in alternative direction with high-degree relation.

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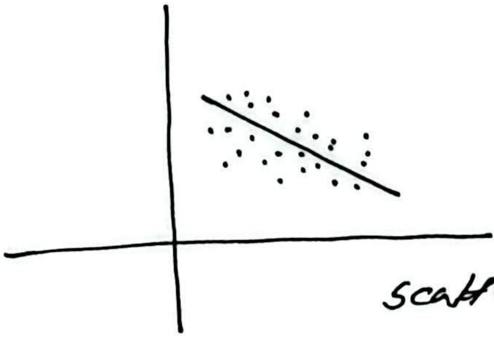
coefficient of determination, $r^2 = 0.64$


scatter diagram.

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1. Section – G

$r^2 = 0.81$, the applied method can determine 81% of the variation of provided data.

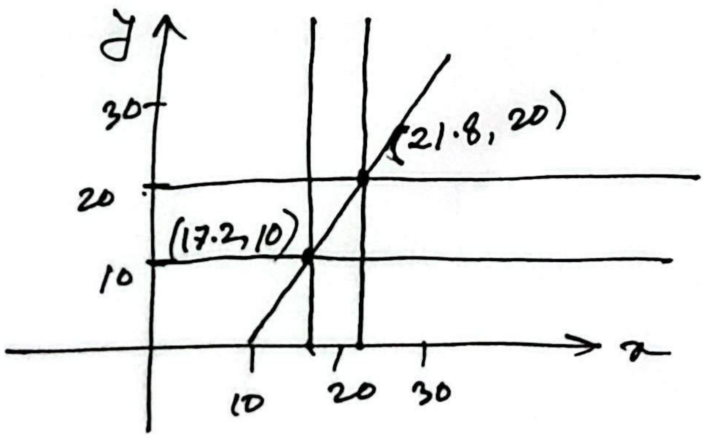
$b_{y/x} = -2.31$, the variable y will decrease by 2.31 for the increase of x by 1.

2. $r^2 = b_{y/x} \times b_{x/y} \Rightarrow (0.65)^2 = 0.92 \times b_{x/y}$

$\Rightarrow b_{x/y} = 0.46$

RL, $\hat{x} = 0.46(y - 25.1) + 12.6$

$\Rightarrow \hat{x} = 0.46y + 1.05$



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3. c_1: 3 1 4 5 2

c_2: 4 3 1 2 5

d_i: -1 -2 3 3 -3

d_i^2: 1 4 9 9 9 → 32

$$\rho = 1 - \frac{6 \times 32}{5(25-1)}$$
$$= -0.6$$

4. $r = -0.75$

scatter diagram.

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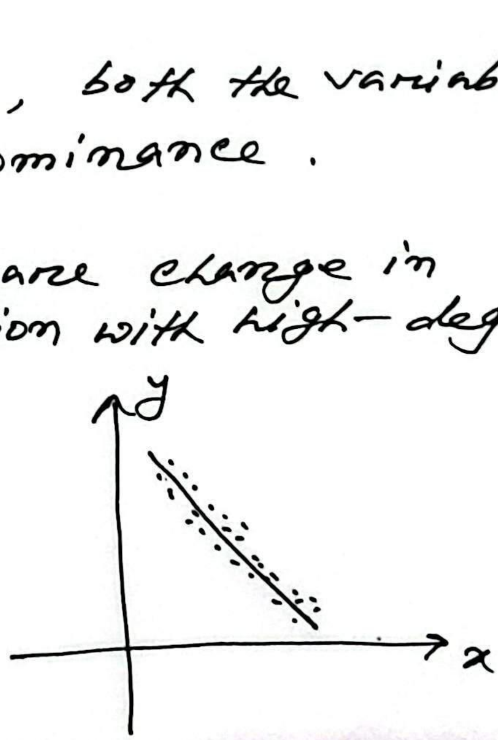
section—0

1. (i) $b_{y/x} = -0.973$, $b_{x/y} = -0.973$

$$r = -\sqrt{(-0.973)(-0.973)} = -0.973$$

(ii) As $b_{y/x} = b_{x/y}$, both the variables have the same dominance.

(iii) The variables are change in alternative direction with high-degree relation.



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2. $r^2 = 0.35$, the applied method can determine 35% of the variation of provided data.

3.

R_1:	3	4	5	2	1
R_2:	3	1	2	5	4
d_i:	0	3	3	−3	−3
d_i^2:	0	9	9	9	9 → 36

$$\rho = 1 - \frac{6 \times 36}{5(25-1)}$$
$$= -0.8$$

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Section–B

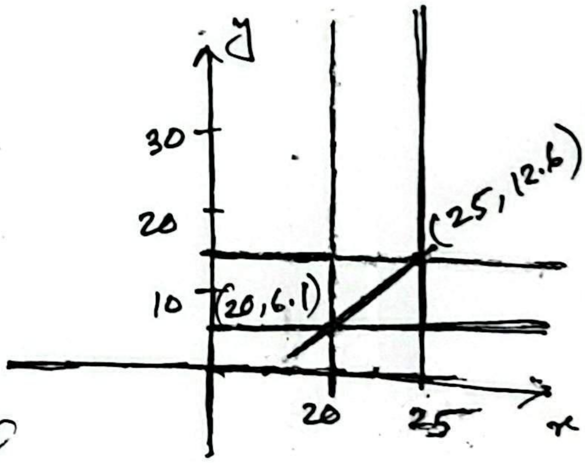
1. (i) $b_{y/x} = r\left(\frac{S_y}{S_x}\right) = 0.85\left(\frac{2.31}{1.52}\right) = 1.29$

The variable y will increase by 1.29 for the increase of x by 1.

(ii) $\hat{y} = 1.29(x - 27.1) + 15.3$

$\Rightarrow \hat{y} = 1.29x - 19.7$

(iii) $\hat{y}(30) = 19$



2.

According to the given diagram the coefficient of regression will be negative. so, the variables will be changed alternatively